UZH

Universität Zürich

ADVANCED MACHINE LEARNING 2026 · GROUP 05

Child Vocalisation Context Classification

A cross-linguistic ACQDIV benchmark across three under-resourced languages

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Do young children produce consistent vocalisations according to the context?

- **Studies show consistency in what the early cries mean**

  Wermke (2025): distress levels affect melody
  Lee et al. (2016) influence of the native language, specially after 6 months

- **From 12 months onwards, we know that children acquire words of their native language**

  First word acquired at 1 year old (Bollinger, 2002)

- **What happens in between?**

  Our question: can the interactional context of a vocalisation be recovered from its acoustics alone — and if so, how strong is that signal against a chance floor?

DATASET

1,009	9	3	1.33 s
audio clips	children	languages	mean clip length

Language coverage

Shipibo-Conibo	Peru	**3 children**
Tagalog	Philippines	**3 children**
Vietnamese	Vietnam	**3 children**

PROBLEM SETTING

4-class classification, cross-speaker generalisation

Formulation

Each example

$$(x_i^{audio},\ x_i^{meta},\ y_i)$$

Goal — learn $f_\theta : (x^{audio}, x^{meta}) \to y$

that generalises to an UNSEEN child speaking a language unseen at validation time.

Metric: macro-F1 (unweighted mean of per-class F1)

Label space $y \in$ { PLTOY, MOVE, INTER, EXTEV }

PLTOY	Playing with a toy	**240**
MOVE	Movement activity	**222**
INTER	Social interaction	**325**
EXTEV	External event	**222**

Note. *Class imbalance is mild (INTER 32%); we report macro-F1 rather than accuracy precisely because INTER's frequency would let a majority-class baseline look misleadingly strong.*

EVALUATION PROTOCOL

Participant-disjoint holdout · 5 seeds · macro-F1

Split	Children	Clips	Language
Train	7 children	569 (56.4%)	Shipibo · Tagalog · Vietnamese
Validation	1 (child_2)	171 (16.9%)	Shipibo-Conibo
Test	1 (child_6)	269 (26.7%)	Tagalog

- The goal is to measure performance equally across all four interactional contexts regardless of their frequency.
- **A uniform-random baseline (sampling from the label distribution) is reported as a statistical reference floor.**

- **Splits fixed once**
- **5-seed variance**

Cross-linguistic confound

Validation child speaks Shipibo (22.5% INTER in training); test child speaks Tagalog.

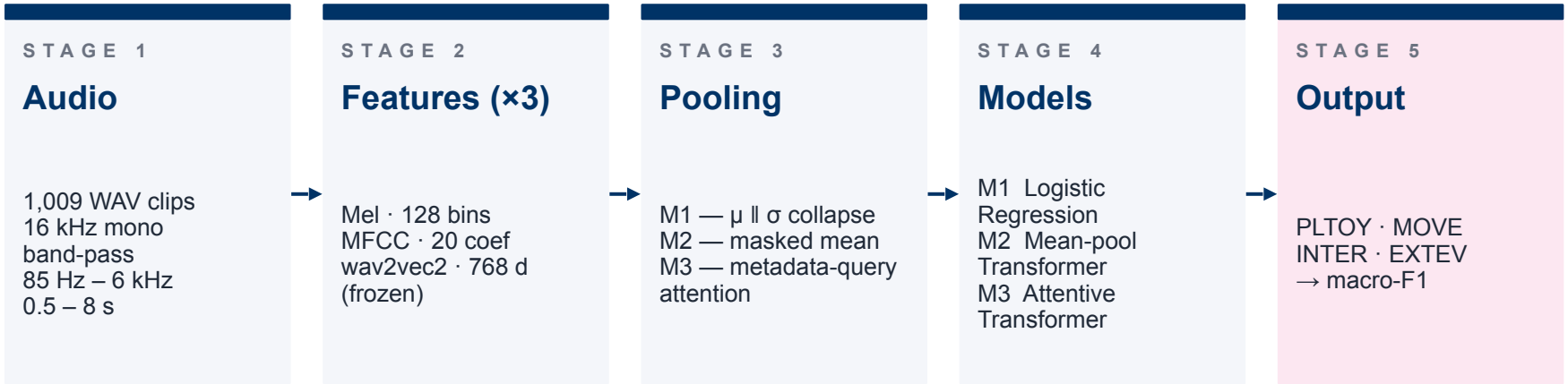
The Tagalog training prior is 46.9% INTER, far above the test child's true 29.4% — this prior/actual gap drives the INTER over-prediction.

The distributional shift between validation and test is structural and cannot be eliminated at this dataset size.

Note. *We did not adopt k-fold CV as the primary protocol because with only 9 children fold composition would dominate variance; LOCO-CV is reported as a supplementary stress-test.*

PIPELINE

From raw WAV to four-class context prediction



Shared preprocessing → identical evaluation protocol → all three models compared on the same clips.

From raw recording to model-ready features

TIER 1 — AUDIO DENOISING *shared by all feature sources*

Raw WAV

anonymised recordings
1,009 clips · 0.5 – 8 s
mono · loaded at 16 kHz

→

Frequency analysis

per-file FFT +
aggregate Welch PSD
across all 1,009 clips
→ choose cutoffs

→

Butterworth band-pass

high-pass 85 Hz · low-pass 6 kHz
order 5 · zero-phase (sosfiltfilt)
forward + backward → no time shift

Removes DC offset + rumble < 85 Hz · removes hiss > 6 kHz · preserves the speech band

TIER 2 — THREE FEATURE SOURCES *one shared downstream protocol (all features × all models)*

Mel spectrogram `128 × 230`

128-bin log-mel · cached PNG
n_fft 2048 · hop 512 · 0–8 kHz

MFCC `20 × 230`

20 coefficients
compact · robust on tiny data

wav2vec2-base `768 × 230`

frozen last-layer embeddings
self-supervised · 960 h LS

SpecAugment (2 × freq masks F=27, 2 × time masks T=70) applied during training on Mel & MFCC only — *not on wav2vec2, where masking would destroy learned phonetic content.*

MODEL 1

Logistic Regression — the simple baseline

1. Time-axis collapse x = (μ(X) ‖ σ(X))
2. StandardScaler (fit on train only)
3. L2-regularised softmax · C tuned via inner LOPO-CV
4. class_weight = balanced · lbfgs (deterministic)

Properties

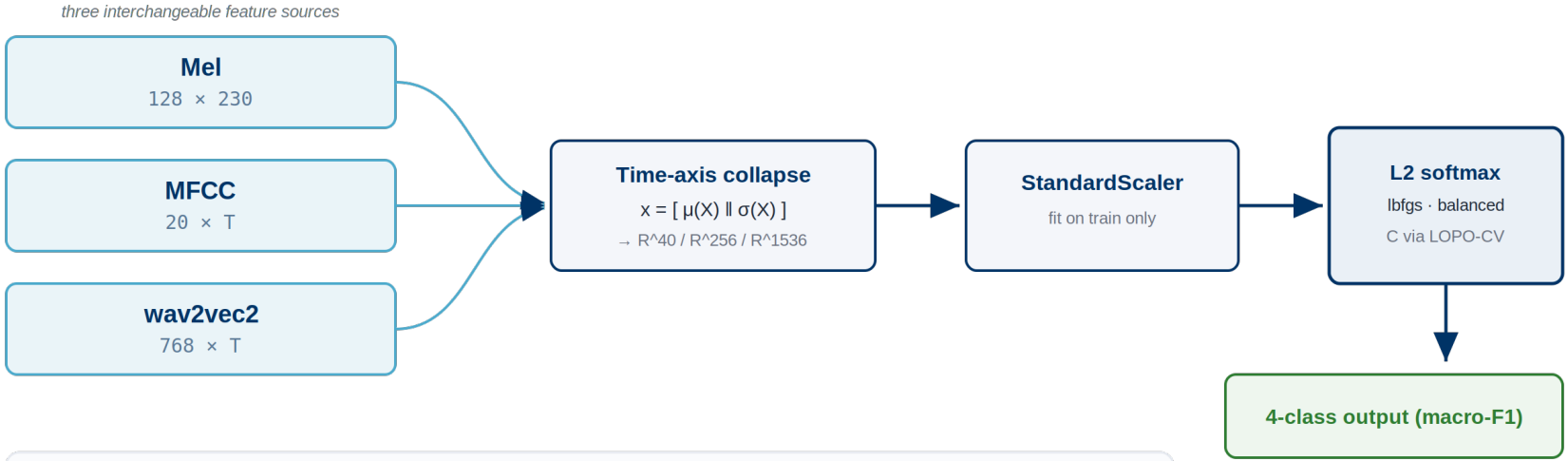
- Deterministic — solver converges to a unique optimum
- Seed variance = 0 (no random init, no stochasticity)
- Direct interpretability — class-conditioned weights

Parameter counts

MFCC	164 params
Mel	**1,028 params**
wav2vec2	6,148 params

MODEL 1 — ARCHITECTURE

Logistic Regression: collapse → scale → softmax



Parameters

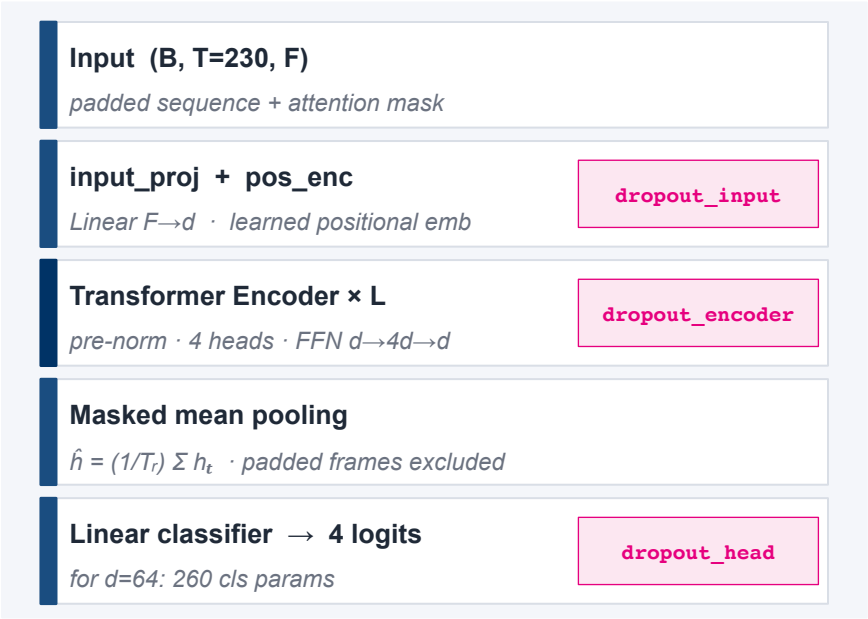
`Mel 1,028 · MFCC 164 · wav2vec2 6,148`

Determinism

lbfgs converges to a unique optimum → seed variance = 0

MODEL 2

Mean-pool Transformer — neural baseline



Design choices

- Per-clip z-score normalisation across all time–freq bins
- Pre-norm Transformer (more stable on small data than post-norm)
- Masked pooling — padded frames receive zero weight
- v2: three independent dropout locations (was: single param)

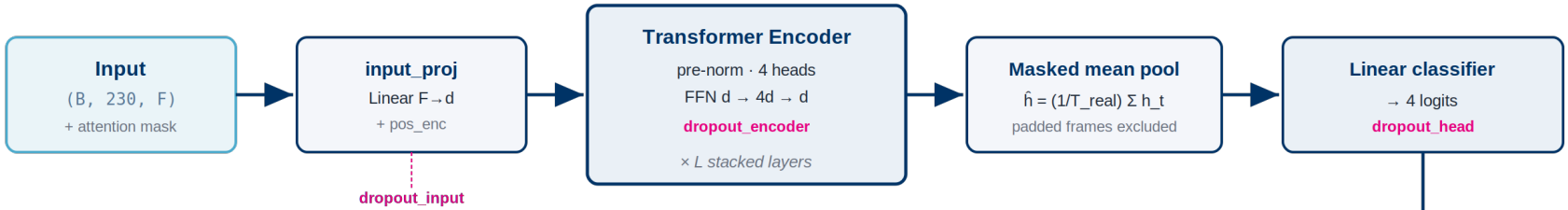
Canonical Mel config

d=64 · L=2 · dropout=0.5 · lr=1e^{-3}
Test macro-F1:
0.249 ± 0.020 on Mel · 5 seeds

Note. *Mean pooling is intentionally weak — keeping a metadata-free baseline lets us isolate Model 3's metadata-query attention as a single architectural change rather than a confounded multi-difference comparison.*

MODEL 2 — ARCHITECTURE

Mean-pool Transformer: encoder → masked mean → classifier



Canonical config (mel)

`d = 64 · L = 2 · lr = 1e-3`

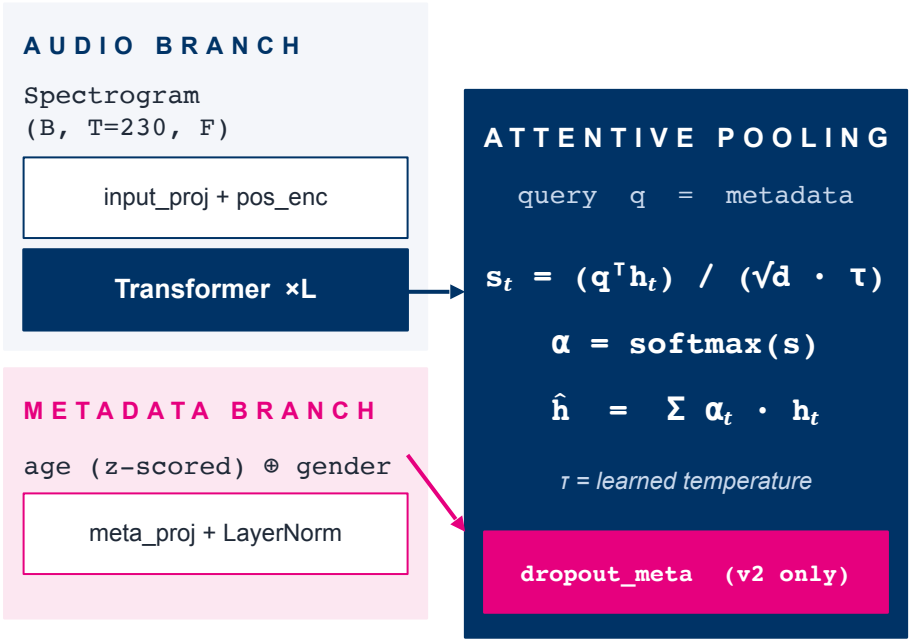
Regularisation (v2)

three dropout sites: input · encoder · head

SpecAugment on mel / MFCC inputs only

MODEL 3 — MAIN MODEL

Metadata-query attentive pooling



What this buys us

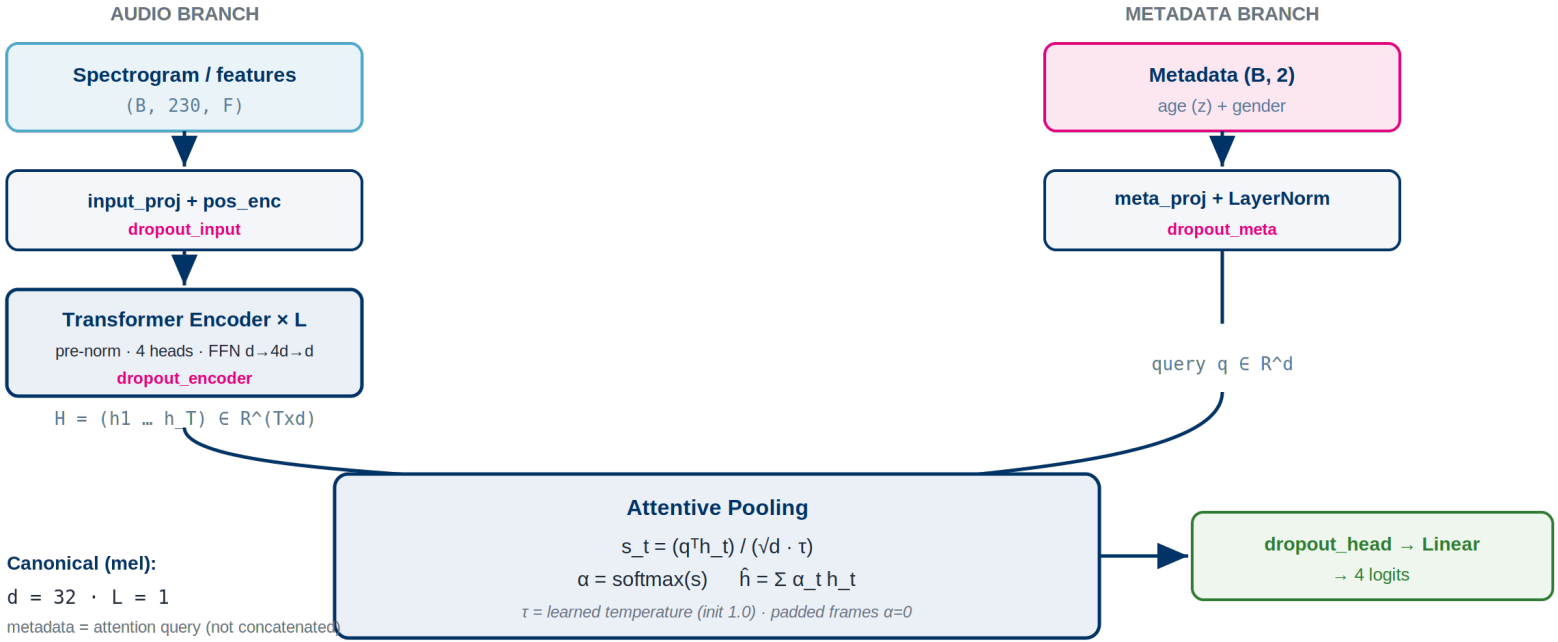
- Metadata never enters the audio stream — the encoder is unchanged from M2
- Attention weights condition pooling on age + gender
- Single architectural change isolates the contribution
- Learned temperature τ controls attention sharpness
- Age carries the metadata signal in comparison to only gender (0.247 vs 0.221)

Canonical Mel config

d=32 · L=1 · dropout=0.3 · lr=1e^{-3}
Test macro-F1:
0.221 ± 0.013 on Mel · 5 seeds

MODEL 3 — ARCHITECTURE

Attentive Transformer: metadata-query pooling over frames



Discrete grid search → canonical results

LR · inner LOPO-CV over 7 train children

9-pt logspace grid

$C \in \{ 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}, 10^{3}, 10^{4} \}$

Test F1 →

MFCC	$C = 10$	**0.253**
Mel	$C = 0.01$	**0.277**
wav2vec2	$C = 0.01$	**0.263**

Transformers · Val macro-F1 selection

8-config grid per (model, feature)

- d_model ∈ { 32, 64 }
- L ∈ { 1, 2 } encoder layers
- dropout ∈ { 0.3, 0.5 }
- lr fixed = 10^{-3} (mel/MFCC) · 5×10^{-4} (wav2vec2)

→ ***2 × 2 × 2 = 8 configurations / cell***

Bayesian optimisation with Optuna TPE

3 × 80	6 × 60	v2 dropouts	9 cells
trials · LR line search $C \in [10^{-6}, 10^{6}]$	trials · Transformer multi-dim search	input · encoder · head + meta (M3 only)	every (model, feature) combination retuned

Transformer search space

```
d_model ∈ {32, 64, 128}   ·   L ∈ [1, 2]
lr ∈ log-uniform [10⁻⁴, 10⁻²]
dropout_input · dropout_encoder · dropout_head
dropout_meta  (M3 only)
objective: validation macro-F1
```

Why Optuna over the grid?

- Continuous C captures values between grid points
- TPE allocates trials to promising regions
- Seeded with 9 grid points → never worse than the grid
- Stress-tests the val-F1 selector under language shift

More parameters do NOT improve accuracy

0.311

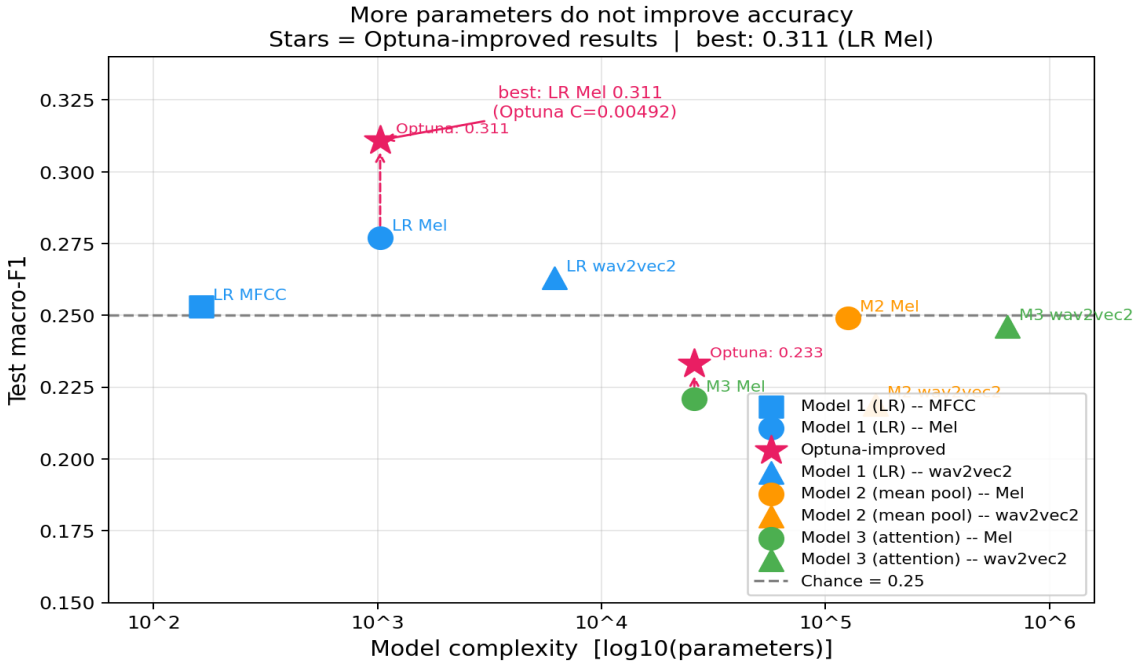
LR · Mel · Optuna C = 0.00492

Best test macro-F1 of the project — 1,028 params

0.251

M2 · wav2vec2 · Optuna

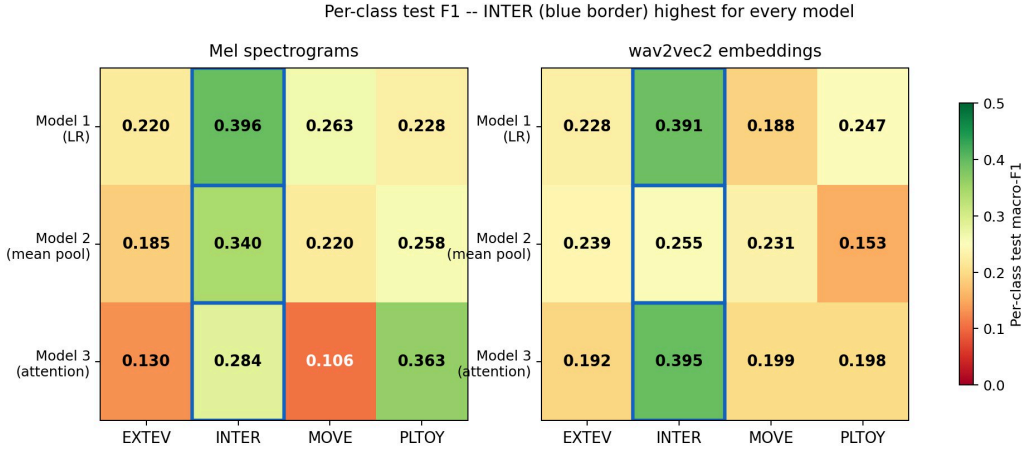
Only one transformer genuinely improves: M2 wav2vec2, 0.219 → 0.251.

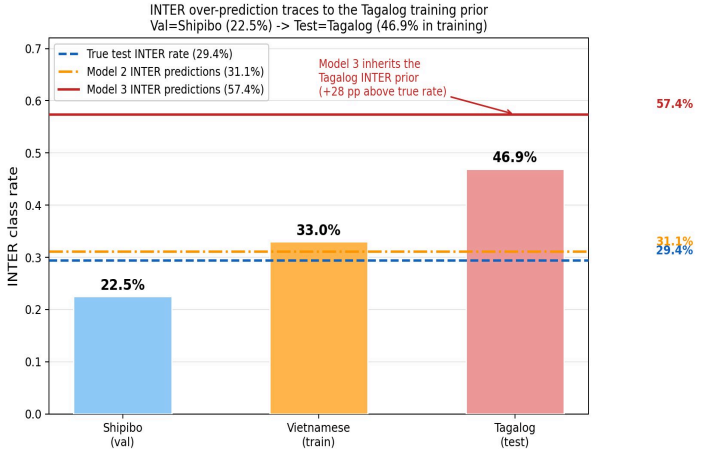


Note. *1,028-parameter LR beats the best canonical attention transformer, which has about 648K parameters.*

PER-CLASS BREAKDOWN

INTER dominates most models: highest in 5/6 panels.

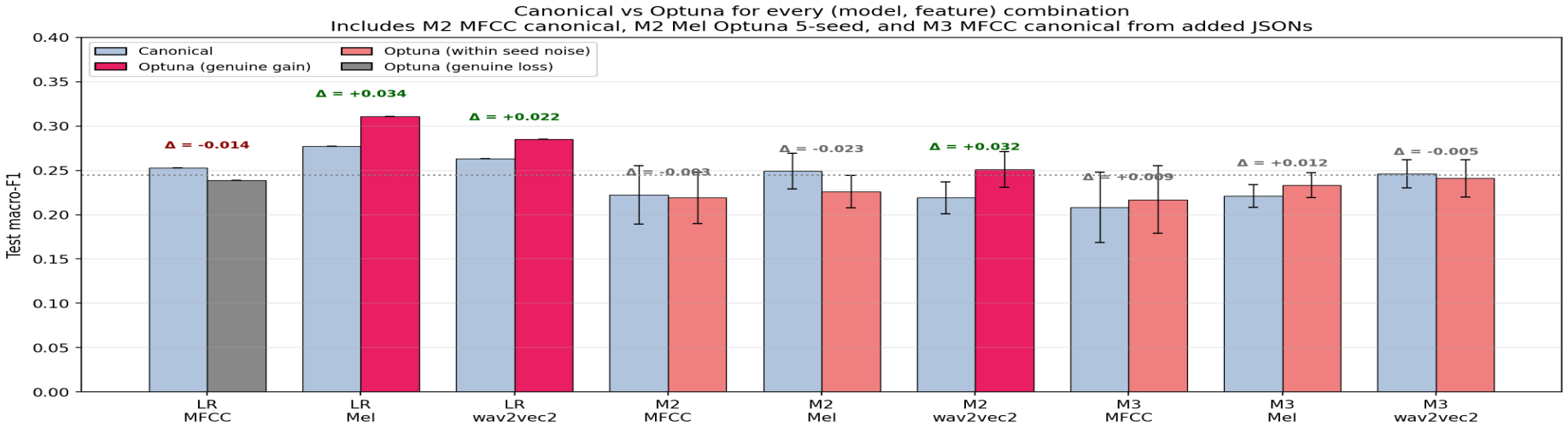




Two compounding effects. (1) Tagalog training prior is 46.9% INTER vs. true test rate of 29.4%. (2) On wav2vec2, M3's attention collapses to a single frame in 95% of clips — most likely interaction-like content. The bias is a distribution-shift artefact, not an architectural failure.

OPTUNA OUTCOMES

Three genuine gains · six cells within seed noise or worse



LR Mel +0.034

Continuous C = 0.00492 lies between grid points 0.001 and 0.01

M2 wav2vec2 +0.032

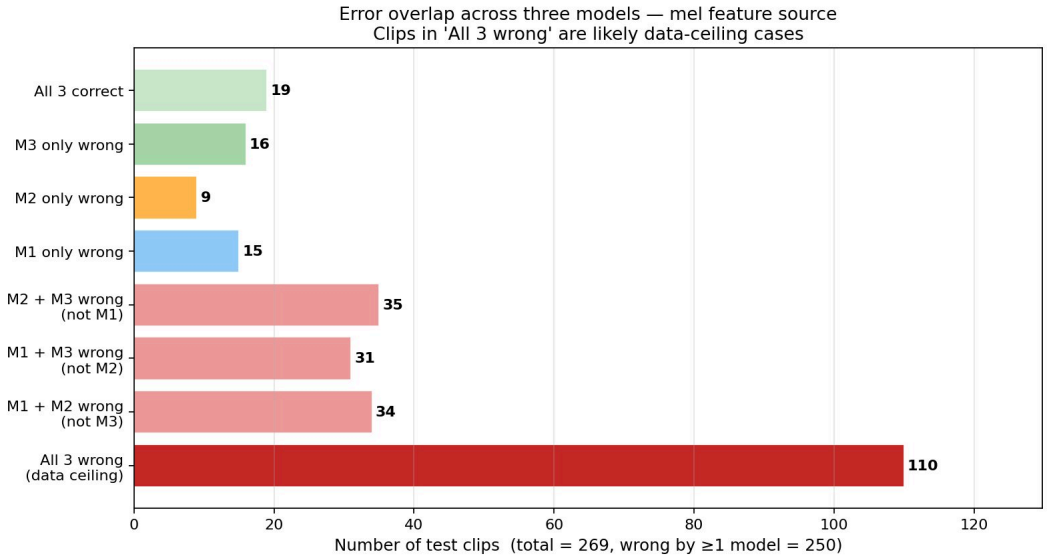
Only Transformer to genuinely beat its canonical baseline. Canonical val-test gap was small → search adds signal, not overfit.

M2 Mel −0.023

M2 Mel Optuna: 0.226 ± 0.018 vs canonical 0.249 ± 0.020 → Δ = −0.023, within seed noise.

Note. *Optuna helps when the canonical val-test gap is small (selector reliable) and hurts when the gap is large (selector fits Shipibo). The pattern across 9 cells is consistent with the language confound, not random noise.*

The ceiling is in the data, not the model



110 clips wrong by all 3 models

19 clips correct by all 3 models

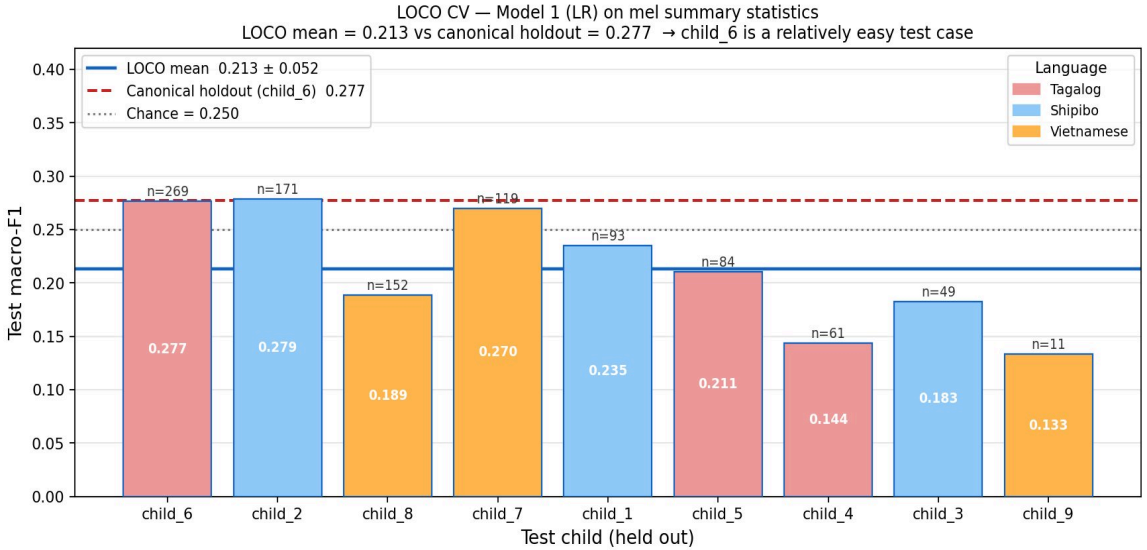
70% Each model's own per-clip error rate

Linear model and transformer baselines fail on the same 110 clips."
Bottleneck is in the data, not the architecture.

Note. *We use this 'all-models-wrong' set as a soft empirical ceiling. Any future architecture that doesn't materially shrink this 110-clip set is unlikely to be improving over LR for the right reason.*

SPEAKER IDENTITY IS THE DOMINANT BOTTLENECK

Leave-One-Child-Out cross-validation · Model 1 mel



Variance decomposition

σ speaker

0.053

LOCO across 9 children

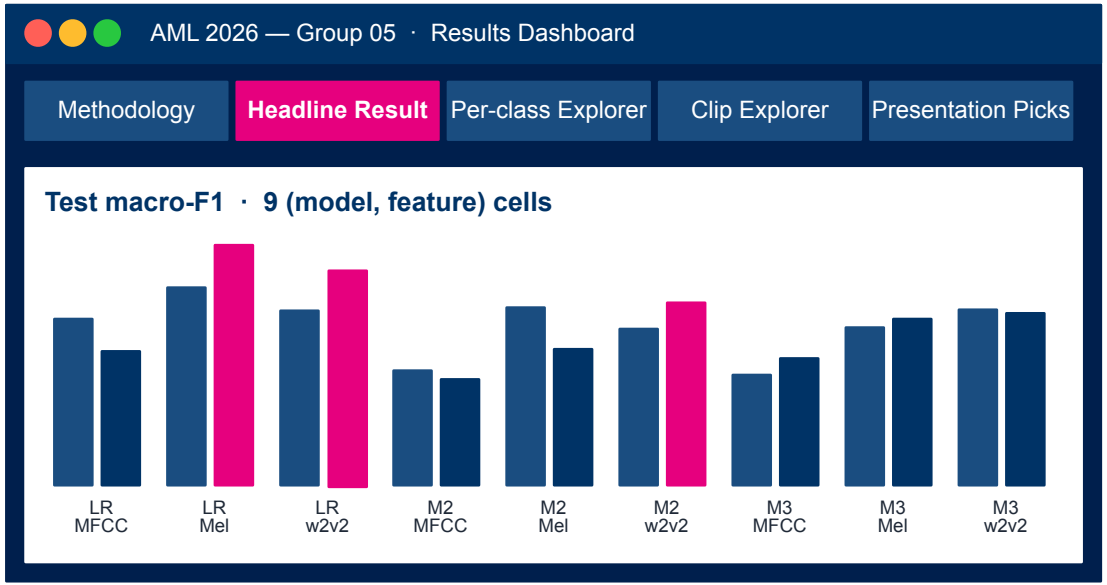
σ seed

0.000

LR is deterministic (lbfgs)

INTERACTIVE DASHBOARD

Reproducible result browser · live demo



Dashboard tabs

Methodology

Pipeline diagram, model cards, training details, hyper parameter tuning — pulled from a single YAML config.

Headline Result

Comparing model complexity vs test F1

Per-class explorer

Per-class breakdown across models and features

Clip explorer and Presentation picks

Our test files and what each model predicted + hand-picked clips that tell the story of the project

CONCLUSIONS

Three findings & a research agenda

01

Simplicity wins at this scale

LR on mel summary statistics (1,028 params) achieves the best test macro-F1 of 0.311 — outperforming a 648K-param attention Transformer by 26%.

02

Three genuine Optuna gains — All where the gap was small

LR mel +0.034, LR wav2vec2 +0.022, M2 wav2vec2 +0.032. Where canonical val-test gap is large, hyperparameter search overfits the Shipibo validation child.

03

Speaker diversity, not capacity, is the bottleneck

$\sigma_{speaker} = 0.053$ vs $\sigma_{seed} = 0.000$. Same 110 test clips fail all three architectures — a clear data-driven ceiling.

FUTURE WORK

Matched-language validation splits · Partial wav2vec2 fine-tuning · LOCO for transformer models · Richer non-identity contextual metadata (recording setting)

Thank you ·